

Scalability & Future Plan

Date	2 July 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited dataset	Prediction accuracy may reduce	High
2	Works only for selected crops	Cannot support all crop types	Medium
3	Basic user interface	Less user-friendly experience	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users	Deploy on cloud to support more users
2	Data Storage	Local storage	Use cloud database for larger datasets
3	Performance	Basic ML model	Optimize model and improve processing speed
4	Security	Basic authentication	Add secure login and data encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add more crop and soil datasets	3 Months	Better prediction accuracy
Phase 3	Develop Android Mobile Application	6 Months	Easy access for farmers
Phase 4	Real-time weather integration and multilingual support	9–12 Months	More accurate recommendations and wider usability